

Assignment No 19

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Domain : AI & ML

College : T.B Girwalkar Polytechnic Ambajogai

Q1.

First 10 Rows:

```
... longitude latitude housing_median_age total_rooms total_bedrooms \
0 -122.23 37.88 41.0 880.0 129.0
1 -122.22 37.86 21.0 7099.0 1106.0
2 -122.24 37.85 52.0 1467.0 190.0
3 -122.25 37.85 52.0 1274.0 235.0
4 -122.25 37.85 52.0 1627.0 280.0
5 -122.25 37.85 52.0 919.0 213.0
6 -122.25 37.84 52.0 2535.0 489.0
7 -122.25 37.84 52.0 3104.0 687.0
8 -122.26 37.84 42.0 2555.0 665.0
9 -122.25 37.84 52.0 3549.0 707.0
```

```
population households median_income median_house_value ocean_proximity
0 322.0 126.0 8.3252 452600.0 NEAR BAY
1 2401.0 1138.0 8.3014 358500.0 NEAR BAY
2 496.0 177.0 7.2574 352100.0 NEAR BAY
3 558.0 219.0 5.6431 341300.0 NEAR BAY
4 565.0 259.0 3.8462 342200.0 NEAR BAY
5 413.0 193.0 4.0368 269700.0 NEAR BAY
6 1094.0 514.0 3.6591 299200.0 NEAR BAY
7 1157.0 647.0 3.1200 241400.0 NEAR BAY
8 1206.0 595.0 2.0804 226700.0 NEAR BAY
9 1551.0 714.0 3.6912 261100.0 NEAR BAY
```

Last 5 Rows:

```
longitude latitude housing_median_age total_rooms total_bedrooms \
20635 -121.09 39.48 25.0 1665.0 374.0
20636 -121.21 39.49 18.0 697.0 150.0
20637 -121.22 39.43 17.0 2254.0 485.0
20638 -121.32 39.43 18.0 1860.0 409.0
20639 -121.24 39.37 16.0 2785.0 616.0
```



Q2.

```
... longitude latitude housing_median_age total_rooms total_bedrooms \
0 -122.23 37.88 41.0 880.0 129.0
1 -122.22 37.86 21.0 7099.0 1106.0
2 -122.24 37.85 52.0 1467.0 190.0
3 -122.25 37.85 52.0 1274.0 235.0
4 -122.25 37.85 52.0 1627.0 280.0
5 -122.25 37.85 52.0 919.0 213.0
6 -122.25 37.84 52.0 2535.0 489.0
7 -122.25 37.84 52.0 3104.0 687.0
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population households median_income median_house_value ocean_proximity
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3 558.0 219.0 5.6431 341300.0 NEAR BAY
4 565.0 259.0 3.8462 342200.0 NEAR BAY
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7 1157.0 647.0 3.1200 241400.0 NEAR BAY
8 1206.0 595.0 2.0804 226700.0 NEAR BAY
9 1551.0 714.0 3.6912 261100.0 NEAR BAY
```

Explain

I selected the California Housing Dataset because it is a well-known regression dataset used in machine learning. It contains information about housing, population, income, and location in different areas of California

Q3.

```
Missing Values Summary:
...      Missing Values  Percentage (%)
longitude          0         0.000000
latitude           0         0.000000
housing_median_age  0         0.000000
total_rooms         0         0.000000
total_bedrooms     207        1.002907
population          0         0.000000
households          0         0.000000
median_income       0         0.000000
median_house_value  0         0.000000
ocean_proximity     0         0.000000

Missing Values After Handling:
longitude          0
latitude           0
housing_median_age  0
total_rooms         0
total_bedrooms     0
population          0
households          0
median_income       0
median_house_value  0
ocean_proximity     0
dtype: int64

Number of Duplicate Rows: 0
Shape after removing duplicates: (20640, 10)
/tmp/ipykernel_618/3040479127.py:28: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True'
```

Observations

1. The dataset contains **207 missing values** only in the `total_bedrooms` column (approximately **1%** of the dataset).
2. Since the missing percentage is very low, the missing values were **filled using the median** of the `total_bedrooms` column. Median is preferred because it is less affected by extreme values.
3. No missing values were found in any other column.

Q4.

*** Statistical Summary of Numeric Columns:

	longitude	latitude	housing_median_age	total_rooms	\
count	20640.000000	20640.000000	20640.000000	20640.000000	
mean	-119.569704	35.631861	28.639486	2635.763081	
std	2.003532	2.135952	12.585558	2181.615252	
min	-124.350000	32.540000	1.000000	2.000000	
25%	-121.800000	33.930000	18.000000	1447.750000	
50%	-118.490000	34.260000	29.000000	2127.000000	
75%	-118.010000	37.710000	37.000000	3148.000000	
max	-114.310000	41.950000	52.000000	39320.000000	

	total_bedrooms	population	households	median_income	\
count	20433.000000	20640.000000	20640.000000	20640.000000	
mean	537.870553	1425.476744	499.539680	3.870671	
std	421.385070	1132.462122	382.329753	1.899822	
min	1.000000	3.000000	1.000000	0.499900	
25%	296.000000	787.000000	280.000000	2.563400	
50%	435.000000	1166.000000	409.000000	3.534800	
75%	647.000000	1725.000000	605.000000	4.743250	
max	6445.000000	35682.000000	6082.000000	15.000100	

	median_house_value
count	20640.000000
mean	206855.816909
std	115395.615874
min	14999.000000

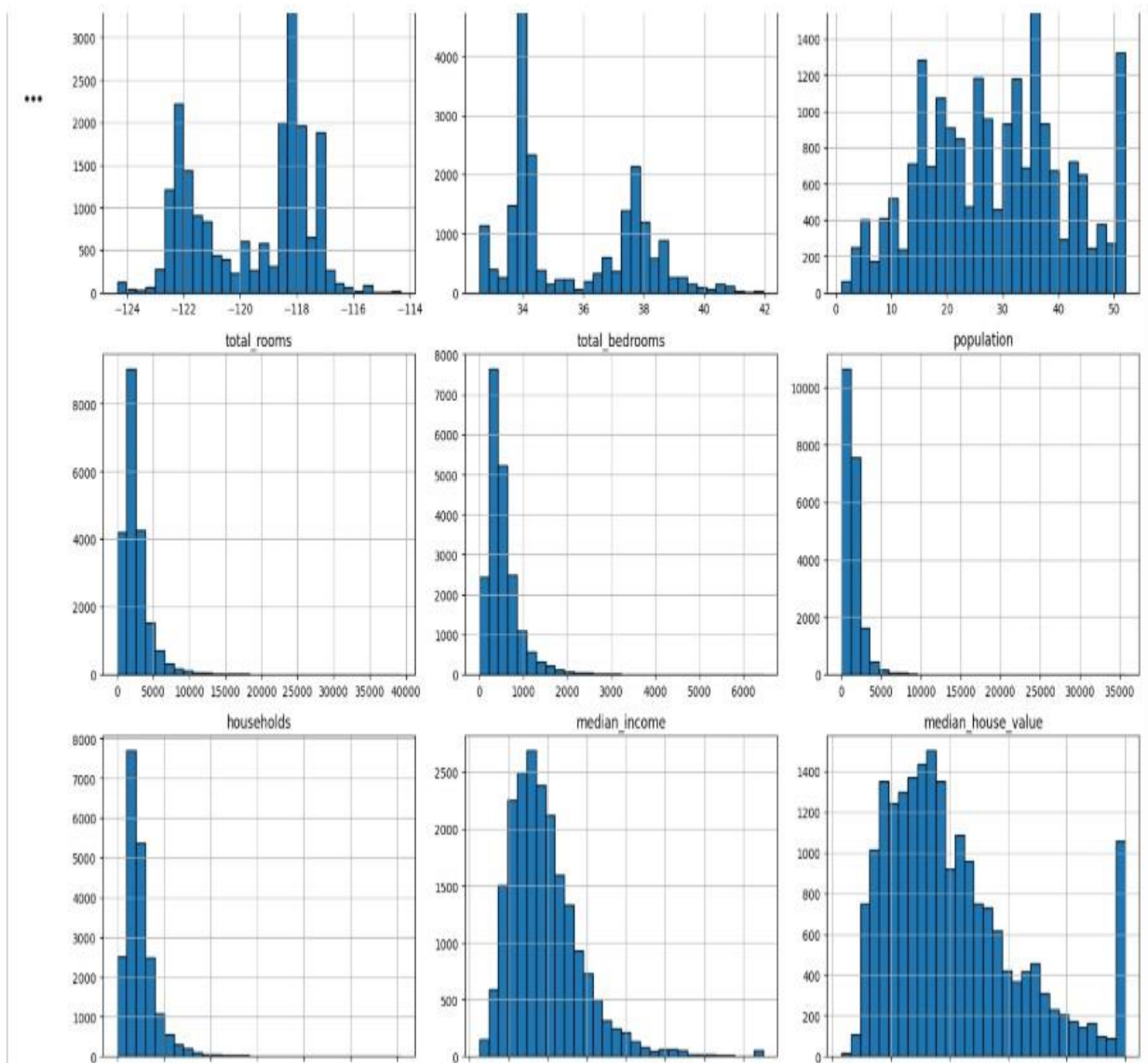
25%	119600.000000
50%	179700.000000
75%	264725.000000
max	500001.000000

Target Variable: median_house_value
Minimum Value : 14999.0
Maximum Value : 500001.0
Mean Value : 206855.81690891474
Median Value : 179700.0

Observations

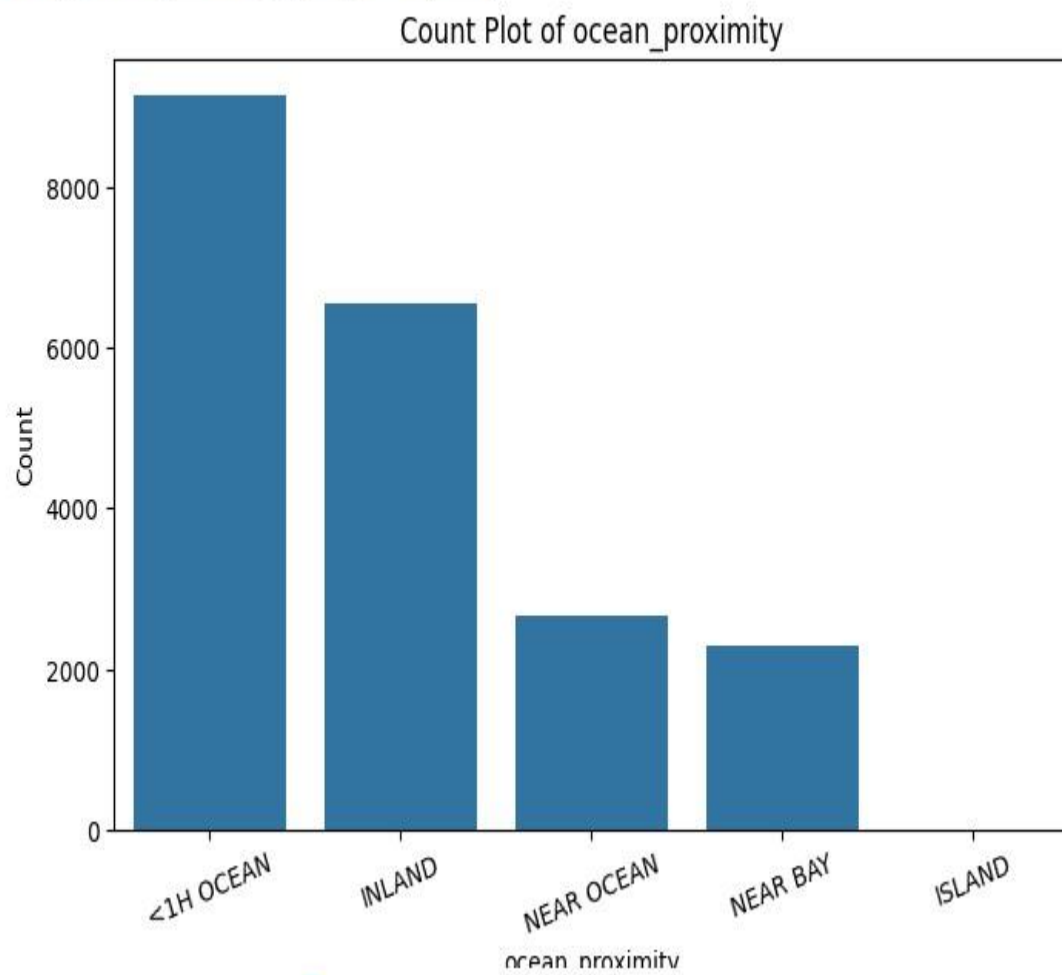
1. The **average house value is approximately \$206,856**, while the **median is \$179,700**. Since the mean is higher than the median, the distribution is **right-skewed**, indicating some high-priced houses increase the average.
2. The **maximum house value is \$500,001**, which is the upper limit in this dataset. This suggests that house prices were **capped** at \$500,001.
3. The **median income** ranges from about **0.5 to 15**, showing a wide variation in household income across different districts.

Q5.



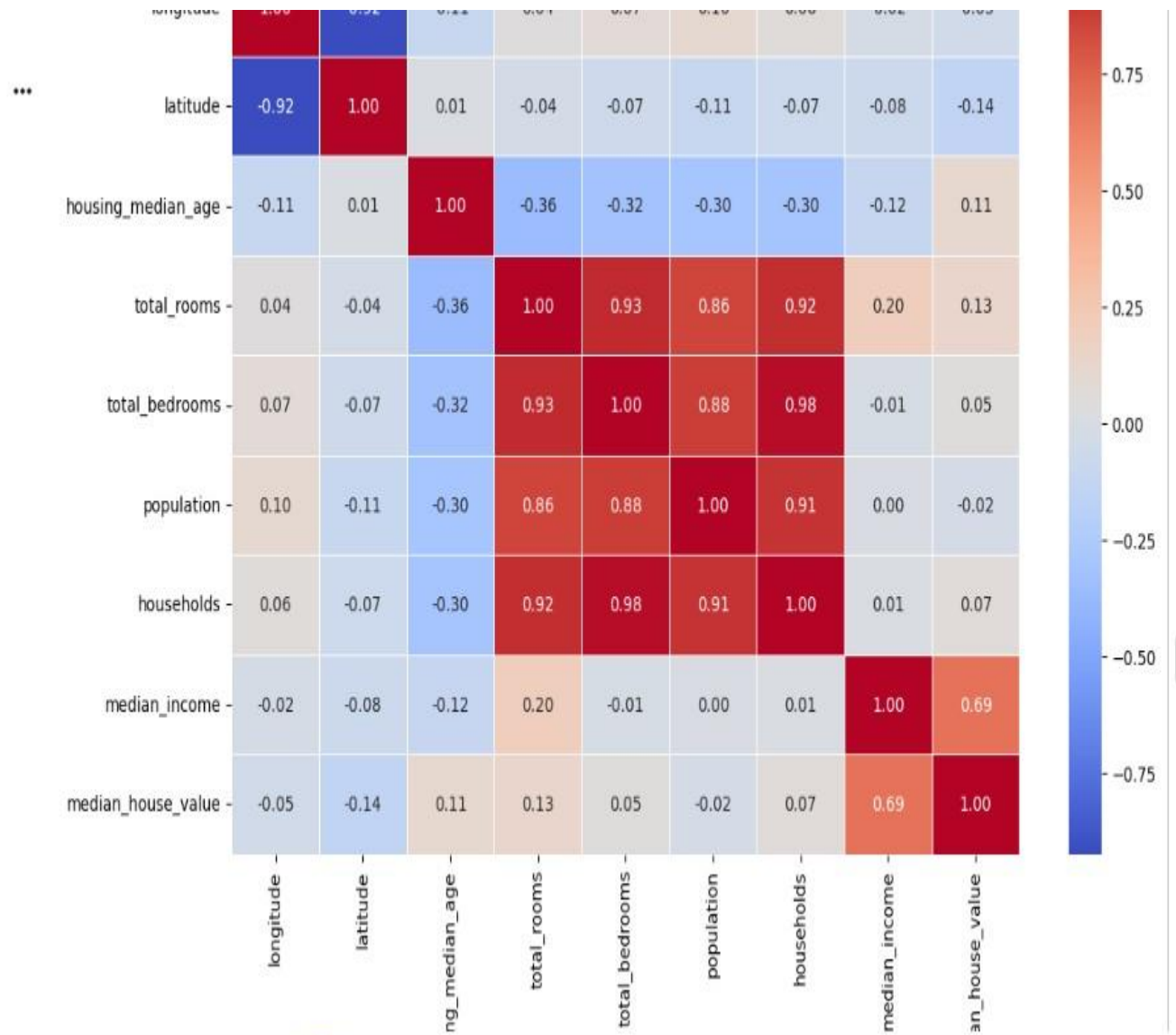
Q6.

```
*** Categorical Columns:  
Index(['ocean_proximity'], dtype='object')
```



Business / Domain Insights

1. **<1H OCEAN** is the **most common category**, indicating that a large portion of California's population lives within one hour of the Pacific Ocean.
2. **INLAND** is the **second-largest category**, representing many residential areas away from the coastline. These locations generally have lower property prices than coastal areas.
3. **NEAR BAY** contains fewer records but represents premium housing markets around the San Francisco Bay Area, where property values are often higher.



Observations

1. **Median income** has the **strongest positive correlation** (≈ 0.69) with **median house value**, indicating that areas with higher household income generally have higher-priced houses.
2. **Latitude** shows a **weak negative correlation** (≈ -0.14) with house prices, suggesting that house values vary across different regions of California.
3. **Total rooms** and **housing median age** have weak positive correlations with the target, meaning they have some influence on house prices but are much less important than median income.

```
✓ ... Independent Features (X):  
    ['longitude', 'latitude', 'housing_median_age', 'total_rooms', 'total_bedrooms', 'population', 'households', '  
  
    Dependent Feature (y):  
    median_house_value
```

Justification

Why are these chosen as Independent Features (X)?

1. These features describe the location, size, age, population, and economic characteristics of each housing block.
2. They provide the information required to estimate the selling price of houses.
3. None of these variables represent the value being predicted; instead, they influence house prices.

Q9.

```
... Categorical Columns:
Index(['ocean_proximity'], dtype='object')
```

```
Before Encoding:
```

```
ocean_proximity
0      NEAR BAY
1      NEAR BAY
2      NEAR BAY
3      NEAR BAY
4      NEAR BAY
5      NEAR BAY
6      NEAR BAY
7      NEAR BAY
8      NEAR BAY
9      NEAR BAY
```

```
After Encoding:
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	\
0	-122.23	37.88	41.0	880.0	129.0	
1	-122.22	37.86	21.0	7099.0	1106.0	
2	-122.24	37.85	52.0	1467.0	190.0	
3	-122.25	37.85	52.0	1274.0	235.0	
4	-122.25	37.85	52.0	1627.0	280.0	
5	-122.25	37.85	52.0	919.0	213.0	
6	-122.25	37.84	52.0	2535.0	489.0	
7	-122.25	37.84	52.0	3104.0	687.0	
8	-122.26	37.84	42.0	2555.0	665.0	
9	-122.25	37.84	52.0	3549.0	707.0	

	population	households	median_income	median_house_value	\
0	322.0	126.0	8.3252	452600.0	

		population	households	median_income	median_house_value \
...	0	322.0	126.0	8.3252	452600.0
	1	2401.0	1138.0	8.3014	358500.0
	2	496.0	177.0	7.2574	352100.0
	3	558.0	219.0	5.6431	341300.0
	4	565.0	259.0	3.8462	342200.0
	5	413.0	193.0	4.0368	269700.0
	6	1094.0	514.0	3.6591	299200.0
	7	1157.0	647.0	3.1200	241400.0
	8	1206.0	595.0	2.0804	226700.0
	9	1551.0	714.0	3.6912	261100.0

	ocean_proximity_<1H OCEAN	ocean_proximity_INLAND	ocean_proximity_ISLAND \
0	0	0	0
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	0	0
7	0	0	0
8	0	0	0
9	0	0	0

	ocean_proximity_NEAR BAY	ocean_proximity_NEAR OCEAN
0	1	0
1	1	0
2	1	0
3	1	0

4	1	0
5	1	0
6	1	0
7	1	0
8	1	0
9	1	0

Encoded Columns:

```
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',  
      'total_bedrooms', 'population', 'households', 'median_income',  
      'median_house_value', 'ocean_proximity_<1H OCEAN',  
      'ocean_proximity_INLAND', 'ocean_proximity_ISLAND',  
      'ocean_proximity_NEAR BAY', 'ocean_proximity_NEAR OCEAN'],  
      dtype='object')
```

Explanation of Encoding Choice

1. The dataset contains one categorical feature (`ocean_proximity`), so One-Hot Encoding is the most suitable encoding method.
2. One-Hot Encoding creates a separate binary column for each category:
 - `ocean_proximity_<1H OCEAN`
 - `ocean_proximity_INLAND`
 - `ocean_proximity_ISLAND`
 - `ocean_proximity_NEAR BAY`

Q10.

First 5 Rows of Scaled Features:

```
✓ ... longitude latitude housing_median_age total_rooms total_bedrooms \
0 -1.327835 1.052548 0.982143 -0.804819 -0.970325
1 -1.322844 1.043185 -0.607019 2.045890 1.348276
2 -1.332827 1.038503 1.856182 -0.535746 -0.825561
3 -1.337818 1.038503 1.856182 -0.624215 -0.718768
4 -1.337818 1.038503 1.856182 -0.462404 -0.611974

population households median_income ocean_proximity_<1H OCEAN \
0 -0.974429 -0.977033 2.344766 -0.891156
1 0.861439 1.669961 2.332238 -0.891156
2 -0.820777 -0.843637 1.782699 -0.891156
3 -0.766028 -0.733781 0.932968 -0.891156
4 -0.759847 -0.629157 -0.012881 -0.891156

ocean_proximity_INLAND ocean_proximity_ISLAND ocean_proximity_NEAR BAY \
0 -0.681889 -0.015566 2.830742
1 -0.681889 -0.015566 2.830742
2 -0.681889 -0.015566 2.830742
3 -0.681889 -0.015566 2.830742
4 -0.681889 -0.015566 2.830742

ocean_proximity_NEAR OCEAN
0 -0.384466
1 -0.384466
2 -0.384466
3 -0.384466
4 -0.384466
```